Experimental assessment of a gas-conserving one-dimensional model for entrapped-air-driven geysering

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Abstract

Entrapped-air release through water-filled shafts can cause destructive geysering in stormwater and combined-sewer systems. Network-scale assessment requires a reduced model, but many one-dimensional formulations prescribe gas release and cannot predict the transition from benign venting to eruption. This study evaluates a gas-conserving mixed-flow framework, developed in a companion methods paper, against three published experimental campaigns. The vertical shaft uses the same decoupled two-fluid equations as the horizontal conduit and connects through a regime-conditioned junction closure. Simulations use reported initial conditions and a frozen numerical configuration, with neither prescribed release histories nor fitted geyser heights. In the ventilation-shaft experiments, the model selects the correct branch for both a wide, non-geysering tower and a narrow, geysering tower. It reproduces the principal pressure levels and water-column motions. In the narrow tower, however, the computed interface velocity is approximately one half of the measured value, and interface breakthrough occurs after the experimental event despite an early arrival at the shaft. In the horizontal-

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pipe/vertical-riser campaign, the model classifies 5 of 7 filmed cases and 39 of 63 configurations in the published parameter sweep. All 24 sweep errors are missed geysers near the transition band; no false geyser is predicted. In the junction-chamber campaign, the model reproduces the A2/B3 outcome reversal caused by the downstream boundary and captures the timing and scale of the first C9 transient, while under-resolving the short B3 pressure peak. The comparisons show that resolved gas transport recovers the principal branch physics across distinct apparatuses. Quantitative errors concentrate in shaft-mouth exchange, coherent-film topology, and acoustic/chamber reduction. The framework is therefore useful for mechanism-based system assessment, but its missed-eruption bias must be reduced before stand-alone hazard screening near the regime boundary.

Keywords: geysering, entrapped air pocket, stormwater system, vertical shaft, two-fluid model, transient mixed flow

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1. Introduction

Geysering is the rapid expulsion of an air–water mixture through a manhole, dropshaft, or ventilation shaft during transient filling of a drainage system. The resulting jet can displace covers, damage nearby infrastructure, and endanger the public. Field pressure records rule out a purely inertial water-column explanation because the piezometric head near an erupting shaft need not reach ground level [1]. Controlled experiments instead identify the release and expansion of a large entrapped air pocket through a partly or fully water-filled shaft as the principal driving mechanism [2, 1].

Laboratory evidence now spans several manifestations of this mechanism.

Vasconcelos and Wright [2] isolated air-pocket release in a single ventilation shaft and documented both benign venting and geysering. Cong et al. [3] varied riser diameter and initial pocket volume systematically and proposed a two-parameter regime criterion. Later studies examined larger shafts [4], prototype-motivated junction chambers [5], covered manholes [6], and mitigation measures [7]. Related multiphase-flow work has also isolated the rapid expansion of a Taylor bubble in a vertical liquid column [8]. Together, these studies show that geysering is not set by a pressure threshold alone. It emerges from competition among pocket compression, gas delivery to the shaft, counter-current exchange, and the time available for the water column to reach the rim.

Available numerical approaches resolve different parts of this competition. Three-dimensional volume-of-fluid simulations can reproduce the local interface and jet structure of an individual event [9]. Their cost, however, limits their use for network-scale studies involving many shafts and rainfall scenarios. One-dimensional mixed-flow solvers are more suited to that scale, but their gas representation is often prescribed. Preissmann-slot and two-component-pressure models primarily follow liquid pressurization [10]. Entrapped gas is then introduced through a lumped pocket law, a specified release rate, or a boundary condition. Rigid-column two-phase models likewise require the gas-pocket and water-column topology to be known in advance [2]. Such models can interpret a selected apparatus, but they cannot independently test whether the computed gas–liquid dynamics select venting or eruption.

A companion methods paper [11] addressed this modelling gap for transient mixed flow in closed conduits. Its stratified branch conserves gas mass and gas momentum within a decoupled two-fluid system. Its liquid-full branch uses an elastic-pipe closure, while moving pressurized–stratified interfaces are solved as Regime-Transition Riemann Problems. A conservative cut-cell update tracks these fronts without prescribing their motion. The methods paper established the numerical formulation through canonical mixed-flow tests and included a schematic geyser-onset example. It did not quantitatively assess the framework against geysering experiments.

The present study provides that application-level assessment. The companion framework is extended to each experimental layout by representing the shaft as the same one-dimensional two-fluid system rotated into the vertical. A local junction closure connects the shaft to the horizontal conduit. The closure is conditioned on the resolved flow regime and geometry, but its parameters and the numerical configuration are frozen before comparison. Consequently, gas release, water-column displacement, and regime selection are computed outcomes. No measured release history or geyser trajectory is imposed.

The assessment is organized around three complementary evidence sets. The first campaign compares time-resolved pressure, interface, and free-surface histories for the non-geysering and geysering ventilation-shaft tests of Vasconcelos and Wright [2]. The second treats the model as a regime classifier against the seven filmed riser tests and the 63-configuration parameter sweep of Cong et al. [3]. The third examines the junction-chamber experiments of Liu et al. [5], including a downstream-boundary branch reversal and a transient mass- oscillation event. These campaigns test increasingly complex questions: whether the model selects the correct branch, whether that selection generalizes across a regime map, and whether the reduced representation retains the first-order dynamics of a prototype-motivated chamber.

This paper therefore asks a narrower question than the companion methods study: how far can a gas-conserving one-dimensional framework reproduce observed geysering before unresolved junction, film, and acoustic physics become controlling? Section 2 summarizes the formulation and defines the application-specific closures. Section 3 reports the three experimental comparisons. Section 4 interprets the common successes and failure modes, and Section 5 states the resulting scope of use.

2. Numerical model and application closures

The governing method was developed and verified in the companion paper [11]. The present section defines the subset needed for the experimental assessment and makes the application-specific assumptions explicit. It first states the mixed-flow balance and moving-front treatment. It then describes the vertical-shaft representation, the shaft-mouth closures, and the numerical configuration used in all three campaigns. Detailed derivations and canonical verification remain in [11]; they are not repeated as contributions of this paper.

2.1. Decoupled two-fluid mixed-flow formulation

The model partitions a closed conduit into liquid-full and stratified gas–liquid reaches separated by moving regime fronts. Stratified reaches obey the decoupled two-fluid balance law of [11],

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}, \tag{1}$$

with conservative state, flux, and regular source vectors

$$\mathbf{U} = \begin{bmatrix} A_g \rho_g \\ \rho_g Q_g \\ A_l \\ Q_l \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \rho_g Q_g \\ \rho_g Q_g^2 / A_g + P_g A_g \\ Q_l \\ Q_l^2 / A_l + \frac{1}{2} \Lambda_d A_l^2 \end{bmatrix}, \qquad \mathbf{S} = \begin{bmatrix} 0 \\ P_g \dfrac{\partial A_g}{\partial x} - A_g \rho_g g \cos\theta \dfrac{\partial h_l}{\partial x} + \mathcal{S}_g \\ 0 \\ \mathcal{S}_d \end{bmatrix}, \tag{2}$$

where A_g and A_l are the gas and liquid cross-sectional areas, ρ_g the gas density, Q_g and Q_l the gas and liquid volumetric flow rates, P_g the gas pressure, h_l the liquid depth, θ the pipe inclination, and $\mathcal{S}_g$, $\mathcal{S}_d$ the regular gas- and liquid-momentum sources containing axial gravity, wall friction, and interfacial shear; the shear closures follow standard stratified-flow correlations [12, 13]. Gas mass and gas momentum are conserved variables, so gas transport, gas inertia, gas compressibility, and shear-driven interfacial growth are part of the computed state. The restoring coefficient in the liquid momentum flux is

$$\Lambda_d = \underbrace{\frac{2gH_g}{A_l}}_{\text{gas pressure}} + \underbrace{\frac{\rho_l - \rho_g}{\rho_l} g\zeta}_{\text{hydrostatic–buoyancy}} - \underbrace{\frac{\rho_g (u_g - u_l)^2}{\rho_l A_g}}_{\text{IKH slip}}, \tag{3}$$

which combines the gas-pressure, hydrostatic–buoyancy, and inviscid Kelvin–Helmholtz (IKH) slip contributions; here H_g is the gas pressure head, u_g and u_l the phase velocities, and $\zeta = \cos\theta / [D \sin(\gamma/2)]$ a circular-section geometric factor with D the pipe diameter and γ the liquid central angle.

The eigenvalues of the decoupled liquid flux Jacobian,

$$\lambda^{\pm} = u_l \pm \sqrt{\Lambda_d A_l}, \tag{4}$$

show that the liquid subsystem is strictly hyperbolic for $\Lambda_d > 0$ and reaches the neutral interfacial-instability (IKH) threshold at $\Lambda_d = 0$; beyond it, interfacial waves grow and liquid bridging can occur.

In liquid-full reaches the gas phase is absent and the pressurized branch is governed by the two-variable system

$$\frac{\partial A_l}{\partial t} + \frac{\partial Q_l}{\partial x} = 0, \qquad \frac{\partial Q_l}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q_l^2}{A_l} + gA_f h_s\right) = \mathcal{S}_p, \tag{5}$$

closed by the elastic-pipe relation

$$P - P_{\text{ref}} = \rho_l a^2 \, \frac{A_l - A_f}{A_f}, \qquad h_s = \frac{P - P_{\text{ref}}}{\rho_l g}, \tag{6}$$

with a the (numerical) pressure-wave speed, A_f the undeformed pipe area, h_s the piezometric head, and $\mathcal{S}_p$ the regular pressurized-branch source; this branch recovers the water-hammer limit. In limiting cases the same area-variable form of Eqs. (1)–(6) recovers water hammer, open-channel gravity flow (with $\Lambda_d \to g\cos\theta/B_l$, where B_l is the free-surface top width), interfacial-instability dynamics, and the near-full mixed-flow balance [11].

Moving pressurized–stratified fronts are internal boundaries between the two-variable system (5) and the four-variable system (1). Their local closure and the conservative front-tracking update are summarized in Sections 2.2–2.3, so that the computations reported here can be interpreted and reproduced from the present paper alone; the derivations, the construction of the active-set solver, and the verification of the scheme on canonical benchmarks are given in [11].

Two features distinguish the present application from a prescribed-release geyser model. First, an entrapped pocket is identified as a connected gas region bounded by liquid seals. Its pressure is evaluated from the current mass and volume through the isothermal relation $P_g = m_g RT/V_g$. The calculation therefore does not impose a pocket-volume history. Second, gas momentum is retained in every stratified reach. Gas arrival at the shaft and its exchange with the water column are outputs of the coupled computation, subject to the junction closures below.

2.2. Regime-Transition Riemann closure at moving fronts

At a moving pressurized–stratified front $x = x_\Gamma(t)$ with speed $w_\Gamma = \dot{x}_\Gamma$, the local coordinate is oriented so that the pressurized branch lies on the left and the stratified branch on the right; the opposite case follows by reflection. Only liquid mass and liquid momentum are common to both branches, and the ideal front has zero thickness and stores neither. Integrating the mixed-regime liquid balance over a control volume shrinking onto the front therefore gives the moving Rankine–Hugoniot conditions

$$\begin{gathered} Q_{p,\Gamma} - Q_{f,\Gamma} = w_\Gamma \left(A_{p,\Gamma} - A_{l,\Gamma} \right), \\ \left(\frac{Q_{p,\Gamma}^2}{A_{p,\Gamma}} + g A_f h_{s,\Gamma} \right) - \left(\frac{Q_{f,\Gamma}^2}{A_{l,\Gamma}} + \frac{1}{2} \Lambda_{d,\Gamma} A_{l,\Gamma}^2 \right) = w_\Gamma \left(Q_{p,\Gamma} - Q_{f,\Gamma} \right), \end{gathered} \tag{7}$$

where the subscripts p and f denote the pressurized trace and the liquid projection of the stratified trace at the front, and $h_{s,\Gamma}$ follows from the elastic closure (6). The pressurized side is liquid-full, so gas does not cross the front; the moving-frame gas condition gives

$$u_{g,\Gamma} = w_\Gamma. \tag{8}$$

Because the water-hammer speed a is much larger than mixed-flow front speeds, one pressurized acoustic characteristic always reaches the front, supplying the compatibility relation

$$u_{p,\Gamma} - u_{p,L} + \frac{g}{a}\left(h_{s,\Gamma} - h_{s,L}\right) = 0, \tag{9}$$

with L the adjacent pressurized state. Equations (7)–(9) do not determine all interface unknowns; the remaining information depends on the position of w_Γ relative to the stratified liquid signal speeds $s_\pm = u_{l,R} \pm c_{l,R}$, $c_{l,R} = \sqrt{\Lambda_{d,R} A_R}$, which partition the admissible front speeds into the slow ($w < s_-$), middle ($s_- \le w \le s_+$), and fast ($w > s_+$) active sets.

Because the correct active set is not known before w_Γ is computed, and branch-dependent nonlinear iteration at every front is fragile in transient computations, the closure of [11] is a fixed-cost one-step construction. Substituting the compatibility relation (9) and the mass jump into the momentum jump reduces the closure, for any candidate speed w, to a scalar quadratic for $A_{p,\Gamma}(w)$ and a scalar momentum residual $R(w)$. A finite-area predictor built from the adjacent states,

$$w_0 = \frac{A_{p,L} u_{p,L} - A_R u_{l,R}}{A_{p,L} - A_R}, \tag{10}$$

is projected onto each (slightly shrunken) active-set interval, one closed-form Newton-type residual correction is applied per interval, and the accepted candidate is the one minimizing a penalized criterion combining the Rankine–Hugoniot residual, the distance to its active set, and positivity penalties. No Bernoulli-type energy relation is imposed at the front. The selected trace defines the moving-frame flux pair $\mathbf{G}_{p,\Gamma} = \mathbf{F}_{p,\Gamma} - w_\Gamma \mathbf{U}_{p,\Gamma}$ and $\mathbf{G}_{f,\Gamma} = \mathbf{F}_{f,\Gamma} - w_\Gamma \mathbf{U}_{f,\Gamma}$ passed to the cut-cell update below.

2.3. Finite-volume update and cut-cell front tracking

Away from fronts, both branches are advanced by the same explicit finite-volume machinery: MUSCL–TVD primitive reconstruction with the minmod limiter, branch-dependent Rusanov fluxes (scalar dissipation $\max(|u_p|+a)$ on the pressurized branch; block-diagonal dissipation with gas speed $|u_g|+c_g$ and stratified liquid speed $|u_l|+c_\Lambda$, $c_\Lambda = \sqrt{\max(\Lambda_d A_l, 0) + c_{\text{num}}^2}$, on the stratified branch), unsplit cell-centered sources, and SSP–RK2 time integration. The time step is the minimum of the branch CFL limits and a front-motion restriction $\Delta t_\Gamma = \frac{1}{2}\min(\Delta x_p, \Delta x_f)/|w_\Gamma^n|$, which prevents the front from sweeping more than half a cell per step.

The cell hosting a front is divided into pressurized and stratified sub-cells of lengths ℓ_p and ℓ_f with $\ell_p + \ell_f = \Delta x$. At the beginning of each step the adjacent branch states are passed to the closure of Section 2.2; the front is then advanced as $x_\Gamma^{n+1} = x_\Gamma^n + w_\Gamma \Delta t$, and each sub-cell is updated by its own branch flux at the outer face and the moving-frame interface flux at the front. Because both sub-cell balances use the same w_Γ and the same interface trace, the interface fluxes and swept-volume contributions cancel exactly for the conserved liquid components and gas mass: the front neither creates nor destroys liquid or gas. When the front crosses a cell face, the affected cells are remapped conservatively (event-based remap), and a bounded, localized residual relaxation corrects the remaining interface mismatch after an accepted step. Trapped gas regions are identified after each step as connected gas volumes bounded by liquid seals, and their pressures are evaluated from the isothermal equation of state on the current pocket mass and volume.

2.4. Vertical-shaft branch

A vertical shaft (ventilation tower, riser, or dropshaft barrel) is discretized as a separate one-dimensional finite-volume branch using the same area–momentum variables as the horizontal conduit, with the pipe inclination set to $\theta = 90°$ so that gravity acts axially. The shaft top is open to the atmosphere; the free surface inside the shaft and any gas core penetrating below it are represented by the axial profile of the gas volume fraction $\alpha_g(z)$, not by a prescribed bubble shape. The shaft state therefore distinguishes the free-surface elevation Y_{fs} (top of the continuous liquid column) from the air–water interface elevation Y_{int} (nose of the gas core rising through the column), which are exactly the two trajectories reported in shaft experiments. Geysering is diagnosed, following the experimental definition of [2], when Y_{fs} reaches the shaft top before Y_{int} catches the free surface.

2.5. Junction and local-loss closures

The shaft base is coupled to the horizontal conduit through a junction closure driven by the local pipe state and the trapped-pocket pressure. The junction exchange uses the pocket equation-of-state static head as the driving head, with the following local-loss elements retained from the frozen configuration used for all runs in Section 3:

- a junction entry/exit loss coefficient $K_j = 0.75$ on the through-flow between pipe and shaft base;
- an additional churn loss $K_g = 8.0$ applied while the junction carries gas, representing the strongly dissipative counter-current air-up/water-down exchange ("glugging") at the shaft mouth;

- a butterfly-valve loss $K_v = 2.0$ at the release valve, whose disc remains in the flow after opening, with a finite hand-opening time $t_v = 0.25$ s represented by throttling the loss coefficient during opening;

- a counter-current flooding (CCFL) limit of Wallis type on reverse liquid drainage at the shaft top while gas is venting, with limiting velocity scale $0.5\sqrt{gD_t}$, where D_t is the shaft diameter.

Three further elements govern the shaft base in the driven regime, i.e. when the trapped pocket holds an overpressure margin over the shaft column ($H_{a0} > Y_{fs,0}$) and gas has reached the junction; they are stated here and motivated with the narrow-tower test in Section 3.1.2: (i) the shaft base is driven by the pipe *crown* pressure (the elevation at which the shaft physically taps the pipe), one bore below the invert piezometric line while the junction is water-sealed; (ii) the shaft gas core connected to the pocket is pocket-fed—mouth influx raises the core nose, by the same fill-to-body-level closure used for the crown cavity in the horizontal pipe—rather than accumulating at the base cell under dispersed-bubble drag; and (iii) the entry gas flux is bounded by the Wallis counter-current flooding scale $j_g \leq C_w^2\sqrt{gD_t\,\Delta\rho/\rho_l}$ with $C_w = 0.9$. These closures use geometric facts and literature-scale constants; in the undriven regime (Test A of Campaign 1) they are inactive and the base retains the invert-datum coupling, a dual treatment whose consolidation is discussed in Section 4.

Once a trapped pocket becomes connected to the shaft and gas breaks through the liquid seal, venting latches on: the pocket continues to discharge through the shaft and is not re-sealed by intermittent slugs. This latch reflects the experimental observation that after breakthrough the release proceeds as a sustained, if oscillatory, discharge rather than a sequence of independent seal events.

2.6. Numerical configuration

All computations in Section 3 use the same numerical settings: $\Delta s = 0.02$ m in horizontal reaches, $\Delta z = 0.01$ m in shafts, and a CFL number of 0.35. The liquid-full branch uses a reduced elastic wave speed $a = 28$ m/s. This value separates the acoustic scale from the approximately 1 m/s gravity scale while limiting grid-scale ringing. It is adequate for pocket-compression and water-column motion, but it is not intended to predict the shortest water-hammer peaks. Gas thermodynamics is isothermal with $R = 287\ \mathrm{J\,kg^{-1}\,K^{-1}}$ and $T = 293$ K, and the liquid density is $\rho_l = 998\ \mathrm{kg\,m^{-3}}$.

No loss coefficient or numerical parameter is fitted to an individual test. The shaft-base treatment is conditioned on the computed state: the crown-datum, pocket-fed, flooding-limited path is active in the driven regime, and the invert-datum path is active otherwise. This distinction is part of the frozen closure logic. It is reported explicitly because the experiments later show that a unified cross-sectional junction closure is still needed.

3. Experimental assessment

The evaluation is organized as a sequence of test campaigns against published geysering experiments. Within each campaign, individual configurations are selected to span the experimentally observed regimes, and all configurations are computed with the single frozen solver configuration of Section 2.6. Simulations start from the documented experimental initial conditions; the shaft motion, the air release, and the regime selection are outcomes of the computation. No prescribed release rate, scripted geyser height, overflow accumulator, or per-case parameter fitting is used.

Three campaigns are selected from the experimental geysering literature so that each contributes a different kind of evidence. Campaign 1 (Section 3.1) reproduces the water-filled ventilation-shaft experiments of Vasconcelos and Wright [2]—the most widely cited controlled geysering experiments and the standard benchmark for geysering models—and tests the time-resolved histories in both regimes: Test A in the non-geysering branch and Test B in the geysering branch. Campaign 2 (Section 3.2) reproduces the horizontal-pipe/vertical-riser experiments of Cong, Chan and Lee [3], the only campaign that condenses its outcome into an explicit two-parameter geysering criterion; it therefore tests the model as a regime classifier across a systematic parameter sweep rather than in individual histories. Campaign 3 (Section 3.3) addresses the prototype-motivated junction-chamber experiments of Liu, Shao and Zhu [5], which connect the laboratory benchmarks to a field-scale dropshaft geometry. Other experimental studies—the field observations of [1], the larger-scale releases of [4] designed for three-dimensional CFD calibration, the covered-manhole geysers of [6], the mitigation study of [7], and the vertical-pipe violent geysers of [14]—are used as corroborating references rather than as reproduction targets, either because their geometry is not controlled at the level needed for one-dimensional comparison or because their focus (cover dynamics, mitigation devices, flashing-driven eruption) lies outside the entrapped-air release mechanism studied here.

3.1. Campaign 1: water-filled ventilation shafts (Vasconcelos and Wright, 2011)

Vasconcelos and Wright [2] used an apparatus comprising a horizontal acrylic pipe of diameter $D = 0.094$ m with both far ends closed, a ventilation tower of length $L = 0.610$ m and variable diameter D_t mounted on a coupling 3.516 m from the upstream end, and a 0.546-m-long upstream air chamber separated from the water-filled reach by a butterfly valve (Fig. 1). The air chamber was pressurized to an initial gauge head H_{a0}, the tower was partially filled to an initial level $Y_{fs,0}$, and each run started when the valve was opened by hand in less than one second. The pressurized pocket then propagated along the pipe crown toward the tower and vented through the water column inside it. A pressure transducer was located 1.07 m downstream of the valve. The experimental programme varied $D_t \in \{57.1, 44.4, 25.4, 12.7\}$ mm ($D_t/D = 0.607$–0.135), $H_{a0} \in \{0.305, 0.610, 0.915\}$ m, and $Y_{fs,0} \in \{0.254, 0.356, 0.457\}$ m. The controlling variable was found to be the tower diameter: for $D_t/D = 0.607$ the free-surface rise never exceeded 10% of L, whereas for $D_t/D = 0.135$ the free surface reached the tower top in every repetition.

Two configurations are selected to span the two regimes (Table 1). Following [2], results are non-dimensionalized by the tower length and the tower gravity-velocity scale: elevations and pressure heads as $Y^* = Y/L$ and $H^* = H/L$, and time as $T^*_{\mathrm{rel}} = t\sqrt{gD_t}/L$. Geysering is defined as the free surface Y^*_{fs} reaching the tower top ($Y^* = 1$) before the rising air–water interface Y^*_{int} catches the free surface.

The one-dimensional model geometry follows the apparatus exactly: up-

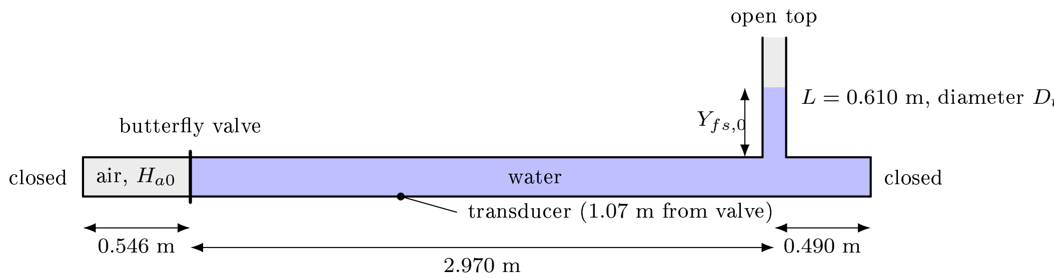

Figure 1: Schematic of the apparatus used by Vasconcelos and Wright [2], as discretized in the one-dimensional model (not to scale in the transverse direction). The horizontal pipe ($D = 0.094$ m) is closed at both far ends; the upstream chamber holds the pressurized air pocket; the ventilation tower (length $L = 0.610$ m, diameter D_t) is open at the top and partially filled to $Y_{fs,0}$.

Table 1: Selected test configurations from the experimental matrix of [2].

Test	D_t (mm)	D_t/D	H_{a0} (m)	$Y_{fs,0}$ (m)	$L/\sqrt{gD_t}$ (s)	Observed regime
A	57.1	0.607	0.305	0.356	0.815	no geysering
B	12.7	0.135	0.610	0.356	1.728	geysering (every run)

stream air chamber 0.546 m (the initial pocket volume, 3.79×10^{-3} m^3), middle pipe 2.970 m, downstream closed pipe 0.490 m, tower at $x = 3.516$ m, and the transducer sampled at $x = 1.616$ m. The initial pocket pressure is $P_{a0} = P_{\text{atm}} + \rho_l g H_{a0}$; the pipe downstream of the valve and the tower below $Y_{fs,0}$ are initially at rest and water-filled. The experimental curves used for comparison were digitized from the published figures of [2] (pressure head from Figs. 5 and 6; free-surface and interface trajectories from Figs. 7 and 8, three experimental repetitions each); the digitized repetitions are shown either individually or as a band.

3.1.1. Test A: large tower, non-geysering branch ($D_t/D = 0.607$)

Test A evaluates whether the one-dimensional model selects the observed non-geysering branch and whether its reduced tower description is consistent with a spatially resolved calculation. The Case-A calculation uses a Tosan-type shock-fitted mixed-flow interface, the conservative true-dry-bed extension, and the pressurized MOC branch also used for Test B; only the Case-A initial gas head and tower parameters are substituted. This horizontal core is coupled to the connected-pocket tower closure. In the experiment, the pocket vents without overflow: the free surface remains near $Y^*_{fs} \simeq 0.6$ during the interface rise and the pressure head retains a long plateau before pocket exhaustion [2]. Both the one-dimensional calculation and the auxiliary two-dimensional OpenFOAM calculation select this branch.

Figure 2 compares three complete-domain states at identical physical times, without a time shift. The rows span the post-impact reflected-bore propagation at $t = 1.50$ s, gas entry into the tower at $t = 6.90$ s, and vented liquid-column recession at $t = 10.10$ s. Both calculations therefore preserve the same non-geysering sequence from horizontal-pocket propagation to tower venting and subsequent drainage. The two-dimensional VOF field additionally resolves lateral bore and interface deformation that the one-dimensional reconstruction cannot represent.

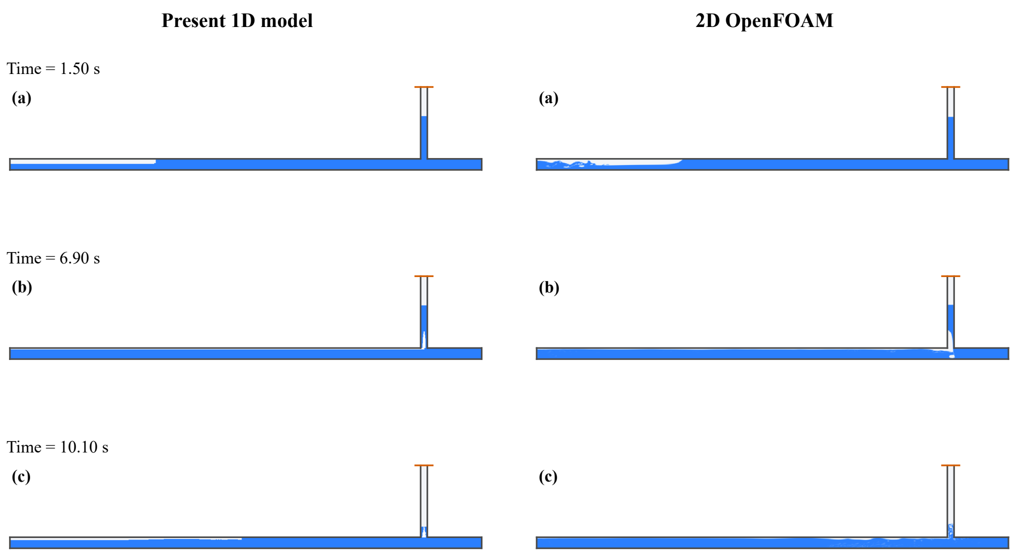

Figure 2: Test A complete-domain phase evolution at identical physical times. Rows (a)–(c) show t = 1.50, 6.90, and 10.10 s, respectively; no time shift is applied. The present one-dimensional model is shown on the left and the two-dimensional OpenFOAM water-volume-fraction field α_{water} on the right. The horizontal pipe, trapped pocket, open pipe–tower junction, and full tower are retained. Both columns use the same pipe-crown datum, coordinate limits, physical aspect ratio, and wall outline. Blue denotes water, white denotes gas, and the red line marks the tower rim. In the one-dimensional reconstruction, the tower gas is displayed as a centred circular core with equivalent diameter $d_g = D_t\sqrt{\alpha_g}$; its plotted width therefore preserves the computed gas-area fraction while leaving the liquid film on both walls.

The unshifted time histories in Fig. 3 provide the quantitative comparison. The contact-preserving release removes the nonphysical start-up spikes seen with the mixed-flow hand-off. Its mean one-dimensional pressure plateau is $H^* = 0.539$ over $4 \leq T^*_{\text{rel}} \leq 7.5$, essentially the digitized experimental median of 0.537. The calculated pressure decrease is nevertheless late: the experiment falls below $H^* = 0.3$ at $T^*_{\text{rel}} = 8.28$, whereas the one-dimensional trace remains near the plateau until the late venting stage. No time or pressure shift is applied. The crown-referenced two-dimensional plateau is 0.562 after conversion from its probe elevation; this is a hydrostatic datum conversion, not a fitted offset.

Panel (b) also removes the artificial flat free-surface history of the previous diagnostic. The one-dimensional free surface rises smoothly during gas entry and reaches $Y^*_{fs,\max} = 0.613$, compared with the largest digitized experimental marker of 0.649 and the two-dimensional maximum of 0.623. The model interface lifts from the tower base at $T^*_{\text{rel}} = 7.85$ and reaches the free surface at 9.18; the corresponding experimental markers begin at 7.29 and meet at about 8.43. Thus the revised calculation reproduces the no-overflow branch, pressure scale, and gradual free-surface rise, while underpredicting the early interface-rise rate and delaying the terminal depressurization. These remaining differences are consistent with representing a coherent gas core and draining wall film by a cross-section-averaged tower closure.

The planar OpenFOAM calculation is used here as a supporting spatial diagnostic, not as a geometry-exact reference solution. In particular, its planar tower-to-pipe area ratio is $D_t/D = 0.607$, whereas the circular experimental apparatus has $A_t/A_p = (D_t/D)^2 = 0.369$; the planar model also cannot reproduce a three-dimensional annular wall film. Accordingly, the experimental curves remain the basis for quantitative validation, while the

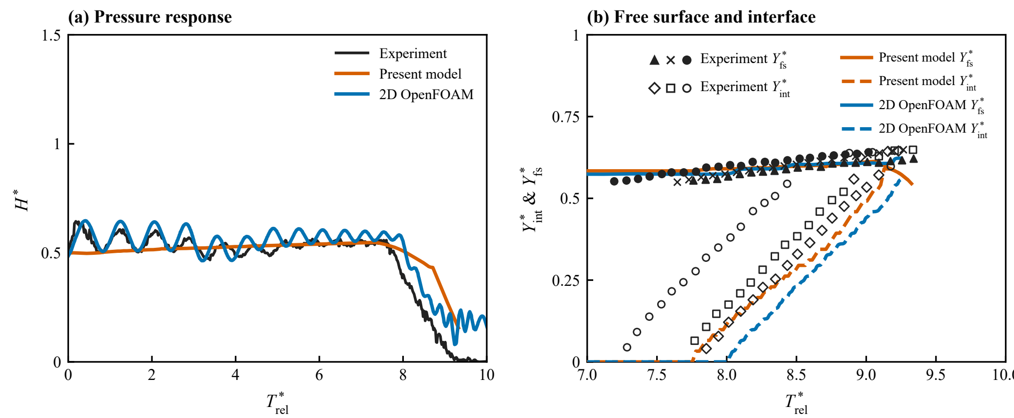

Figure 3: Test A comparison with the digitized measurements of [2], with no time shift. (a) Transducer pressure head: the black line denotes the pointwise median of three experimental repetitions; the present one-dimensional result uses the contact-preserving horizontal-pipe release update and connected-pocket tower closure (red, 0.8 s moving mean), and the crown-referenced two-dimensional OpenFOAM result is shown in blue (0.10 s moving mean). (b) Marker centres digitized from the Case-A panel of the published Fig. 7 are redrawn as equal-sized vector symbols without connecting lines: triangles, crosses, and filled circles identify the three Y^*_{fs} repetitions, whereas open diamonds, squares, and circles identify the three Y^*_{int} repetitions. The corresponding model quantities use solid and dashed curves, respectively; colours identify the two calculations, and each curve stops at the earlier of the experimental-record limit or the loss of a uniquely connected tower column.

two-dimensional frames and histories test whether the reduced model follows a physically consistent phase-evolution pathway.

3.1.2. Test B: small tower, geysering branch ($D_t/D = 0.135$)

This narrow-tower configuration provides the geysering counterpart to Test A. Vasconcelos and Wright reported a geyser in every repetition: the three Fig. 8 records first show gas in the tower over $T^*_{\mathrm{rel}} = 3.648$–$3.742$, while the free surface reaches the rim over $T^*_{\mathrm{rel}} = 3.900$–$3.981$ and therefore precedes gas breakthrough. The credible high-interface markers occur over $T^*_{\mathrm{rel}} = 4.09$–$4.17$. The Fig. 6 transducer record has a pre-release plateau near $H^* = 0.76$ and a collapse envelope of approximately $T^*_{\mathrm{rel}} = 4.02$–$4.18$ [2].

The frozen one-dimensional calculation uses the crown-datum, pocket-fed, flooding-limited junction treatment defined in Section 2.5; no Case-B time alignment or coefficient adjustment is introduced. A planar OpenFOAM calculation is included only as an independent spatial diagnostic. It retains the published pipe lengths and initial heads but uses the area-equivalent slit width $W = D_t^2/D$ and neglects surface tension. It is therefore a controlled two-dimensional surrogate for the release pathway, not a geometry-exact representation of the 12.7-mm circular tower.

Both calculations nevertheless select the observed geysering branch (Fig. 4). In the one-dimensional result the gas interface enters the tower at $T^*_{\mathrm{rel}} = 3.52$ and the free surface reaches the rim at $T^*_{\mathrm{rel}} = 3.70$; the corresponding two-dimensional events occur at $T^*_{\mathrm{rel}} = 3.79$ and 4.11. Thus both preserve the experimental ordering—the water surface spills before the rising gas core catches it—although the two-dimensional surrogate develops the sequence later. The snapshot columns are matched by these physical stages rather than by imposing a common clock, and each panel retains its own unshifted T^*_{rel}.

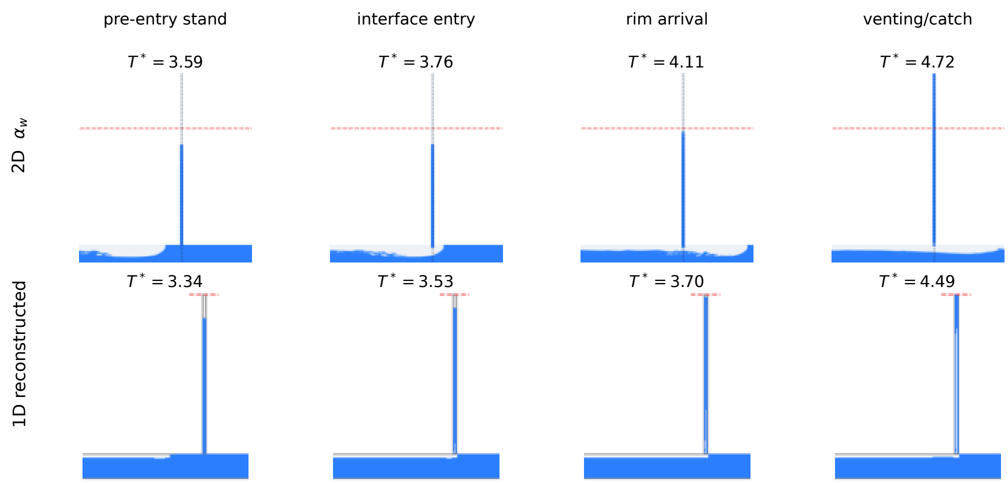

Figure 4: Test B phase evolution in the narrow tower. Top row: OpenFOAM water-volume-fraction field α_w; bottom row: reconstructed one-dimensional phase occupancy. Columns show the pre-entry stand, interface entry, first rim arrival, and venting/catch stage. The red dashed line denotes the tower rim. Columns are stage-matched rather than time-shifted; the value above every panel is that calculation's original T^*_{rel}.

The unshifted histories in Fig. 5 separate bulk-scale agreement from event-timing error. The experimental pre-eruption stand is $Y^*_{fs} = 0.823$, compared with 0.836 in the one-dimensional model and 0.869 in the two-dimensional surrogate. At the crown datum, the one-dimensional pressure plateau is $H^* = 0.754$ versus the digitized experimental value 0.758; the two-dimensional pre-release mean is higher, $H^* = 0.838$ over $1 \leq T^*_{\mathrm{rel}} \leq 3$. The one-dimensional pressure collapse occurs at $T^*_{\mathrm{rel}} = 4.49$, after the experimental collapse interval, whereas the two-dimensional head relaxes more gradually. These comparisons use the native time coordinate throughout.

The isolated digitized point at $(T^*_{\mathrm{rel}}, Y^*) = (3.847, 0.92)$ is retained in the plot for provenance but is not used as an interface-arrival estimate because it lies outside all three rising-interface tracks.

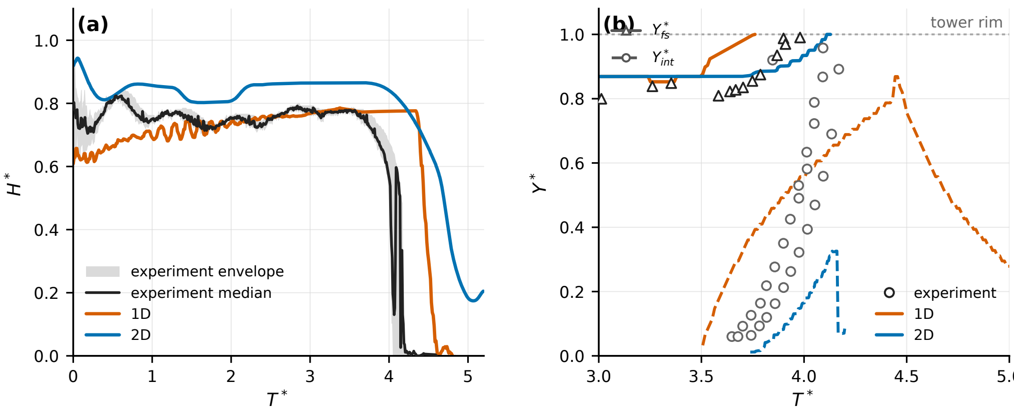

Figure 5: Test B comparison with the centre panels of Figs. 6 and 8 in [2], with no time shift. (a) Crown-referenced pressure head H^*: digitized three-run raster envelope and median (grey/black), one-dimensional result (red, 0.4 s moving mean), and two-dimensional OpenFOAM result (blue, 0.10 s moving mean). The envelope is a dark-pixel span, not a statistical confidence interval. (b) Free-surface elevation Y^*_{fs} (solid; experimental triangles) and gas–water interface Y^*_{int} (dashed; experimental circles). The two-dimensional level series uses a 0.10 s running median and is shown only through the published high-interface window because probe-threshold extraction becomes intermittent after overflow; free-surface traces stop at first rim arrival.

Taken together, Tests A and B show that the frozen, regime-conditioned model selects venting without overflow in the wide tower and eruption in the narrow tower. For Test B the one-dimensional model most closely reproduces the pre-eruption pressure and water-column scales, while its eruption starts early and its blowdown ends late. The planar calculation supports the same entry-to-overflow morphology but overpredicts the pre-release head and water level and delays the principal events. Quantitative validation therefore remains anchored to the experiment; the two-dimensional result tests spatial consistency and exposes the limits of the planar geometric surrogate. Section 4 relates the remaining timing pattern to the shaft-mouth and entrainment closures.

3.2. Campaign 2: entrapped-air release from a horizontal pipe into a vertical riser (Cong, Chan and Lee, 2017)

3.2.1. Experimental apparatus and test conditions

The apparatus of [3] consists of a horizontal acrylic pipe of diameter $D = 0.05$ m and length approximately 6 m, connected at its upstream end to a constant-head tank and carrying a vertical riser of height 1.8 m and variable diameter ($D_r = 16$–46 mm) mounted through a tee-junction. An air pocket of prescribed initial length L_0 is created in the horizontal pipe between ball valves; releasing it under the upstream head H_0 drives the pocket along the pipe crown to the riser, where it vents through the initially water-filled column. Air–water interface trajectories in the pipe and riser were measured by video and high-speed camera, and pressures near the pipe end and at the riser base by transducers. From their parameter study the authors condensed the outcome into a two-parameter geysering criterion,

$$\frac{D_r}{D} \leq 0.62 \qquad \text{and} \qquad V_{air}^* = \frac{V_{air}}{(\pi D_r^2/4)\, H_0} \geq 3.42, \tag{11}$$

i.e. geysering requires both a sufficiently narrow riser and a pocket volume large relative to the riser water column.

This campaign therefore evaluates the model in a different role from Campaign 1: not against individual time histories, but as a regime classifier over a systematic sweep. Two comparisons are made. First, the seven high-speed-camera runs of the experimental Series B (B-H1 – B-H7: $H_0 = 0.66$ m, $L_0 = 0.61$ m, riser diameters $D_r/D = 0.32$–0.92) are computed individually and compared against the measured classification, pocket arrival times, and mean rise velocities. Second, a broader synthetic sweep over the experimental parameter ranges ($D_r = 16$–46 mm, $L_0 = 0.61$–1.8 m, $H_0 = 0.66$–0.88 m; 63 configurations) is computed and the model classification is compared against the experimental criterion, Eq. (11).

All runs of this campaign use the same fully synchronous junction coupling as Campaign 1, with the identical frozen solver configuration (the Campaign-1 Test-B closure set; no parameter is changed between campaigns or between runs). Two geometric elements are specific to the apparatus: the upstream boundary is the constant-head tank, and the pocket is created *downstream* of the riser tee, so the released pocket travels upstream along the crown toward the riser. The tee position (2.88 m from the tank) is fixed from the experiment's own kinematics—the measured arrival times and front speeds of Table 2 of [3] give a valve-to-tee distance $T_a U_f$ of $2.51/1.92/1.32$ m for the three pocket lengths $L_0 = 0.61/1.2/1.8$ m, all placing the tee at the same station—and is held fixed across the sweep. An earlier version of this campaign used a staged horizontal/vertical treatment whose classifier reduced to a static volume–diameter criterion; those results were circular as a validation of Eq. (11) and are superseded by the fully coupled results reported here.

3.2.2. Series B: classification, arrival times, and rise velocities

Table 2 and Fig. 6 summarize the seven Series-B runs. The model reproduces the measured classification in five of the seven: geysering for B-H1 ($D_r/D = 0.32$) and no geysering for B-H4–B-H7 ($D_r/D = 0.62$–0.92), with no false positives. The two intermediate geysering runs B-H2 and B-H3 ($D_r/D = 0.42$, 0.52) are misclassified as non-geysering: the computed free surface is driven to 0.87 and 0.82 m but does not reach the rim. The miss is one-sided—the model errs only toward under-predicting eruption—and its mechanism is identified in Section 4: the volume-neutral mouth exchange, which is what lets the narrow-riser pocket rebuild the overpressure that powers the B-H1 geyser, also cancels part of the one-to-one column lift in the reservoir-fed layout, and the deficit grows with riser width. The computed arrival times are late by 1.5 s on average (largest for the widest riser, where the arrested crown current covers the final reach slowly), against a measured spread of 7.84–8.22 s.

In the geysering run B-H1 the computed eruption-window climb speeds ($v_{fs} = 1.71$, $v_{int} = 1.28$ m/s as the fastest sustained 0.6-s climb) bracket the measured whole-event means (0.92, 1.23 m/s): the eruption jet itself is over-sharp, the same narrow-riser bias identified in Test B of Campaign 1. In the no-geysering branch the computed gas-nose climbs (0.69–1.20 m/s) exceed the measured 0.42–0.48 m/s because the computed climb concentrates in surge pulses rather than the steady Taylor-scale ascent observed; the computed free-surface response (0.20–0.65 m/s during the pulses, zero sustained rise) shows the same pulsed character against the measured steady 0.2–0.26 m/s.

The computed pipe-end pressure transient now differentiates the branches under the synchronous coupling: the geysering case B-H1 develops a compression surge to 1.71 H_0 (cycle-averaged) against about 1.9 H_0 reported, while the no-geysering case B-H6 shows only a gentle post-arrival hump to 1.02 H_0 against about 1.4 H_0 reported. The staged treatment gave both runs the same 2.07 H_0 peak; recovering the experimental differentiation—a wide riser vents the pocket and thereby damps the pipe-end pressure—is the principal gain of the synchronous coupling, obtained at the cost of the two intermediate-diameter classification misses discussed above.

Table 2: Campaign 2, experimental Series B of [3] ($H_0 = 0.66$ m, $L_0 = 0.61$ m): measured against computed classification, pocket arrival time T_a, maximum computed free-surface height $Y_{fs,\max}$ (riser top at 1.8 m), and rise speeds of the free surface (v_{fs}) and gas nose (v_{int}). Measured speeds are whole-event means (Table 2 of [3]); computed speeds are the fastest sustained 0.6-s climb in the venting window. Misclassified runs are marked †.

			Geysering		T_a (s)			v_{fs} (m/s)		v_{int} (m/s)
Run	D_r/D	V^*_{air}	Meas.	Model	Meas.	Model	$Y_{fs,\max}$ (m)	Meas.	Model	Meas. / Model
B-H1	0.32	9.03	yes	yes	8.07	7.56	1.80	0.92	1.71	1.23 / 1.28
B-H2†	0.42	5.24	yes	no	7.84	–	0.87	0.77	–	1.02 / –
B-H3†	0.52	3.42	yes	no	8.18	9.56	0.82	0.66	0.43	0.92 / 1.24
B-H4	0.62	2.40	no	no	8.14	9.40	0.82	0.21	0.53	0.42 / 1.20
B-H5	0.72	1.78	no	no	8.10	9.88	0.86	0.26	0.65	0.48 / 1.05
B-H6	0.82	1.37	no	no	8.10	9.16	0.78	0.25	0.39	0.48 / 1.05
B-H7	0.92	1.09	no	no	8.22	11.44	0.79	0.20	0.20	0.44 / 0.69

Figure 7 shows the two signature runs in detail (these are the runs examined as individual time histories in Sections 3.2.2 and the case folders; the per-run comparisons against the digitized level and pressure histories of [3] Figs. 7, 9 and 10 are reported with the archived reproduction). In B-H1

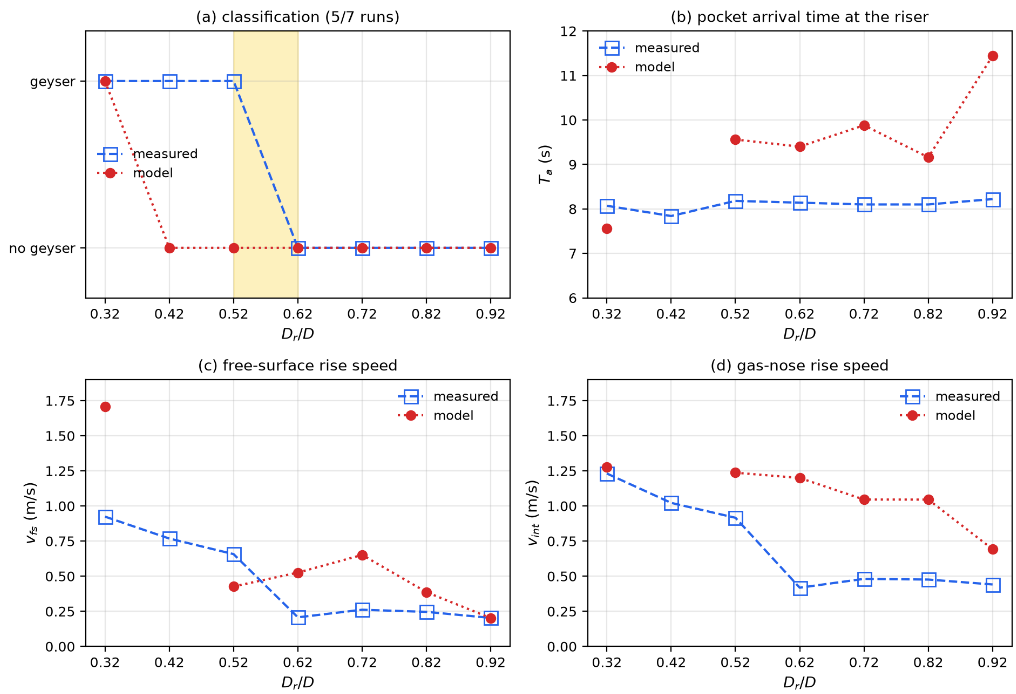

Figure 6: Campaign 2, Series B of [3]: measured against computed (a) geysering classification, (b) pocket arrival time at the riser, (c) mean free-surface rise speed, and (d) mean gas-nose rise speed, as functions of the riser-to-pipe diameter ratio D_r/D.

($D_r/D = 0.32$) the released pocket arrives at 7.56 s (measured 8.07 s), the entering gas drives the free surface up in two surge pulses, and the arrest of the supply-line counter-flow compresses the pocket to 1.71 H_0 (cycle-averaged; reported $\approx 1.9\,H_0$), ejecting the column—the free surface reaches the rim with the gas nose still below it, the geysering signature. In B-H6 ($D_r/D = 0.82$) the same release delivers the same pocket, but the wide riser passes the gas with a gentle push of the free surface to 0.78 m and the pocket vents without eruption; its pressure never exceeds $1.02\,H_0$ after arrival. The branch differentiation of the pipe-end pressure—the experiment's cleanest controlled contrast, since only D_r changes—is reproduced in sign and ranking, though both levels are under-predicted (the $1.4\,H_0$ no-geyser hump is missed because the computed free-surface lift is under-predicted in the reservoir-fed layout, Section 4).

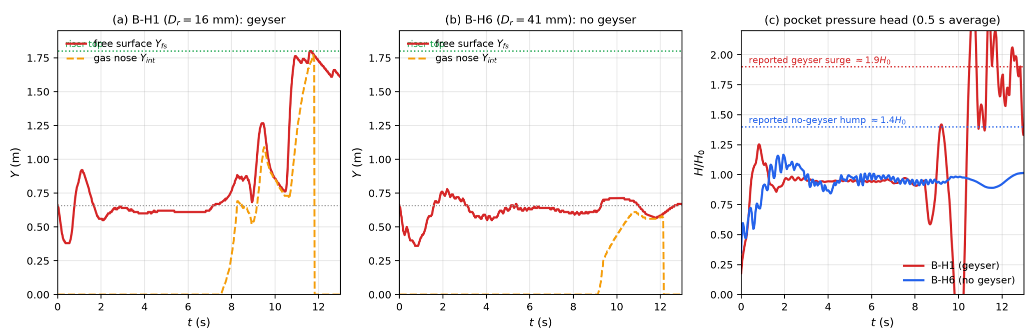

Figure 7: Campaign 2, signature runs of [3] under the fully synchronous coupling: (a) computed riser free-surface and gas-nose trajectories for the geysering run B-H1 (D_r = 16 mm); (b) the same for the no-geysering run B-H6 (D_r = 41 mm); (c) computed pocket pressure head (0.5 s average) for both runs against the reported peak levels ($\approx 1.9\,H_0$ geyser surge; $\approx 1.4\,H_0$ no-geyser hump).

3.2.3. Regime map against the experimental criterion

Figure 8 shows the 63-configuration sweep in the criterion plane $(D_r/D,\ V^*_{air})$. Each configuration is one full synchronous simulation classified with no knowledge of Eq. (11); the criterion lines are drawn for comparison only. The model classification agrees with the experimental criterion in 39 of 63 configurations, and the 24 disagreements are strictly one-sided: in every one the criterion predicts geysering and the model does not. There are no false positives—every configuration the model erupts lies inside the criterion's geysering region, and the no-geysering half-plane $D_r/D > 0.62$ is classified correctly throughout, as is the small-volume region $V^*_{air} < 3.42$. The misses concentrate at intermediate diameters ($D_r/D = 0.42$–0.62) combined with large pocket volumes ($L_0 \geq 1.2$ m), where the computed free surface is driven to 1.0–1.6 m but falls short of the 1.8-m rim; seven of the 24 reach within 0.45 m of it. The pattern is the Series-B under-lift bias extended over the parameter plane: the model captures the physical structure of the regime boundary—both end members and the interaction sense that eruption requires narrow risers *and* large relative pocket volume—but places the eruption threshold conservatively, under-predicting eruption in the transition band. For screening use the bias direction matters: the model as frozen errs toward missing geysers near the boundary, not toward false alarms.

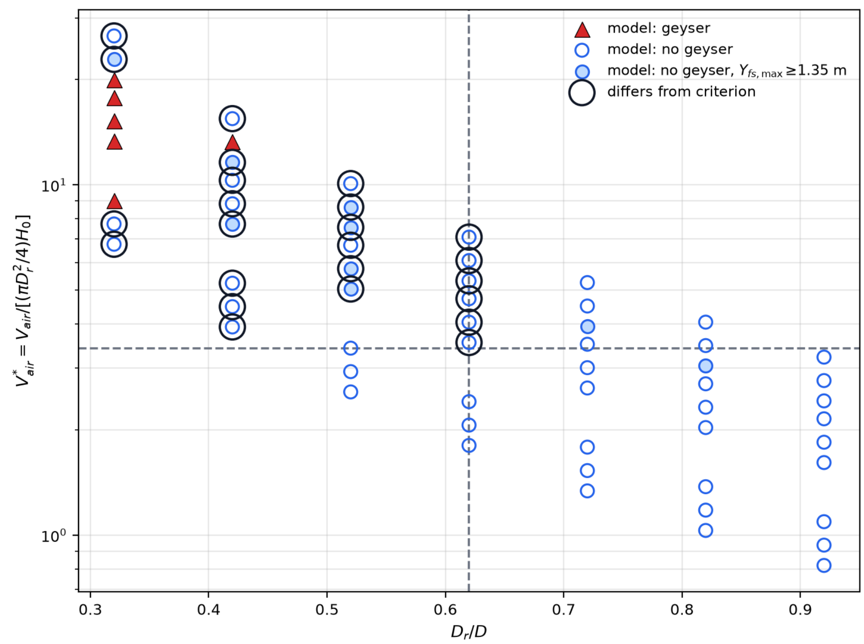

Figure 8: Campaign 2: model geysering classification for 63 configurations spanning the experimental parameter ranges of [3] (D_r = 16–46 mm, L_0 = 0.61–1.8 m, H_0 = 0.66–0.88 m), each a full synchronous simulation, plotted in the criterion plane of Eq. (11). Filled triangles: model geysers; open circles: model does not geyser (filled light circles: free surface within 0.45 m of the rim); black rings: disagreement with the criterion; dashed lines: the criterion thresholds $D_r/D = 0.62$ and $V^*_{air} = 3.42$. The model agrees in 39 of 63 configurations; all 24 disagreements are criterion-geyser/model-no, with no false positives.

3.3. Campaign 3: stormwater geyser in a vertical shaft above a junction chamber (Liu, Shao and Zhu, 2020)

3.3.1. Experimental apparatus and one-dimensional representation

The apparatus of [5] is a 1:20-scale model of a 27 m deep Edmonton prototype dropshaft. A 5.80 m upstream pipe ($D = 0.20$ m, slope 1:100) feeds a transparent junction chamber ($0.30 \times 0.30 \times 0.45$ m) with an invert drop of 0.18 m to a 5.95 m downstream pipe ($D_d = 0.28$ m). A vertical shaft of diameter $d_r = 0.06$ m and height 1.22 m is mounted at the chamber lid centre and vents to the atmosphere. Inflow is imposed as a flow step $Q_0 \to Q_1$ with valve opening time $T_v \approx 0.4$ s. Pressure transducers PT1–PT4 monitor the shaft, the chamber lid (PT2, the primary datum), the chamber floor, and the upstream crown.

Three configurations are computed with the same frozen solver as Campaigns 1–2. The horizontal reach is represented as a single composite one-dimensional domain—upstream pipe, chamber, and downstream pipe—with internal momentum faces at the chamber connections and a rigid-column ordinary differential equation for the shaft water column coupled at the lid tap. This is a reduced one-dimensional formulation: three-dimensional chamber mixing, slug venting at the chamber gas port, and the arrival of a compressing air pocket along the upstream crown in the late-time regime are not resolved at the level of the companion two-fluid/RTRP framework and are stated explicitly where they limit the comparison.

3.3.2. Branch pair A2/B3: downstream boundary flips the outcome

Cases A2 and B3 share the same inflow step ($Q_0 = 20$, $Q_1 = 100$ L/s) and differ only in the downstream tail: A2 discharges to an open channel (weir-controlled, $h_d/D_d = 1/4$), whereas B3 is initially surcharged with a submerged tail gate. In the experiment A2 never geysered while B3 produced a single-shoot eruption on every repetition (Fig. 5 of [5]).

The model reproduces this branch flip without retuning. For A2 the riser column remains below the rim ($h_{r,\max} = 0.47$ m against a shaft height of 1.22 m) and the steady chamber pressures agree with the digitized Fig. 3 record: PT3 5.26 kPa against 4.99 kPa (-5%) and PT2 1.80 kPa against 2.15 kPa ($+16\%$ in the standing-column head that emerges from the composite domain rather than being prescribed). For B3 the computed run geysered, with the free surface reaching the shaft top and overflow. The eruption timing is close—peak PT2 at 1.68 s against 1.47 s—but the compressive peak amplitudes are underpredicted with the frozen elastic branch celerity $a_{wh} = 40$ m/s: PT2 peak 31 kPa against 55 kPa (-44%). A sensitivity run at $a_{wh} = 80$ m/s gives 63.5 kPa, confirming that the deficit scales with the acoustic wave speed rather than with a missing branch-specific closure. The computed jet height $h = 2.17$ m lies on the experimental regression line of Fig. 7(a) ($h = 0.694\,P_{\max}/\rho g + 0.309$ m, $R^2 = 0.97$).

3.3.3. Case C9: transient-driven geysers and the analytic mass-oscillation model

Case C9 ($Q_0 = 25 \rightarrow Q_1 = 40$ L/s, surcharged downstream tail, initial shaft column $h_{r0} = 0.30$ m, sealed air wedge on the upstream crown in the experiment) is the authors' detailed two-stage case: the first two geysers are driven by the hydraulic transient before the trapped pocket reaches the chamber at $t \approx 6.46$ s; geysers 3–8 are pocket-driven and lie outside the present reduced solver.

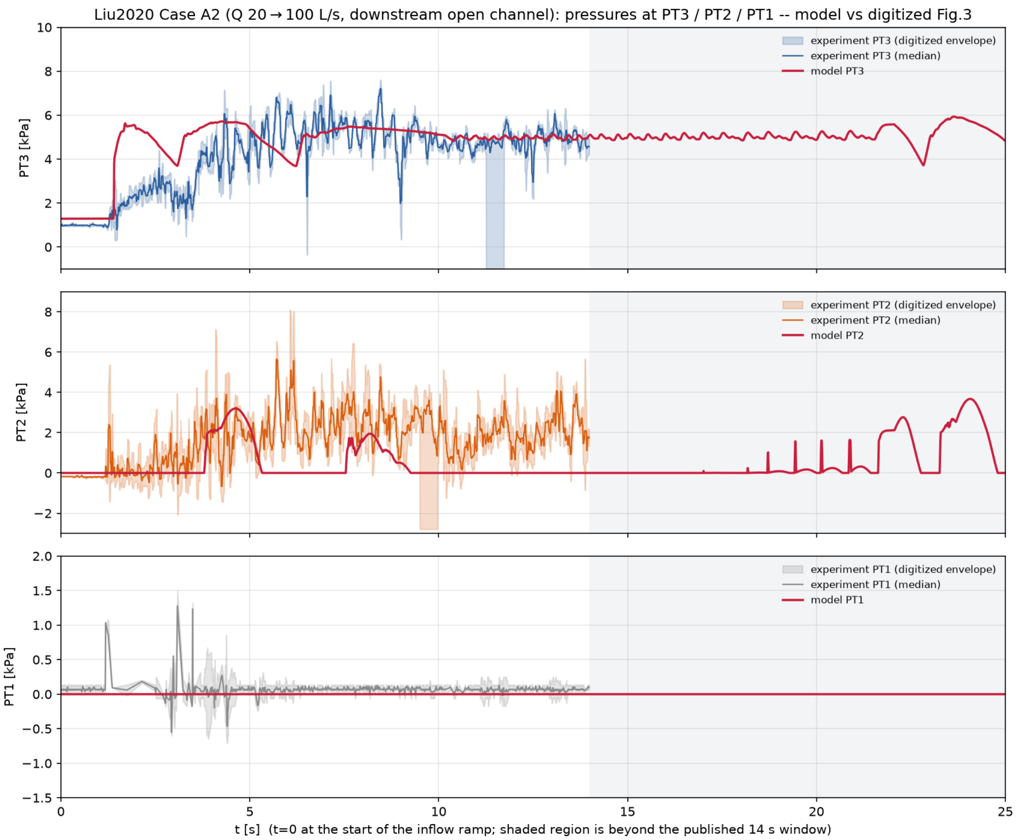

Figure 9: Case A2 (Q: 20→100 L/s, downstream open channel): chamber and shaft pressures against time. Blue: digitized experimental PT2/PT3 from Fig. 3 of [5]; red: present model.

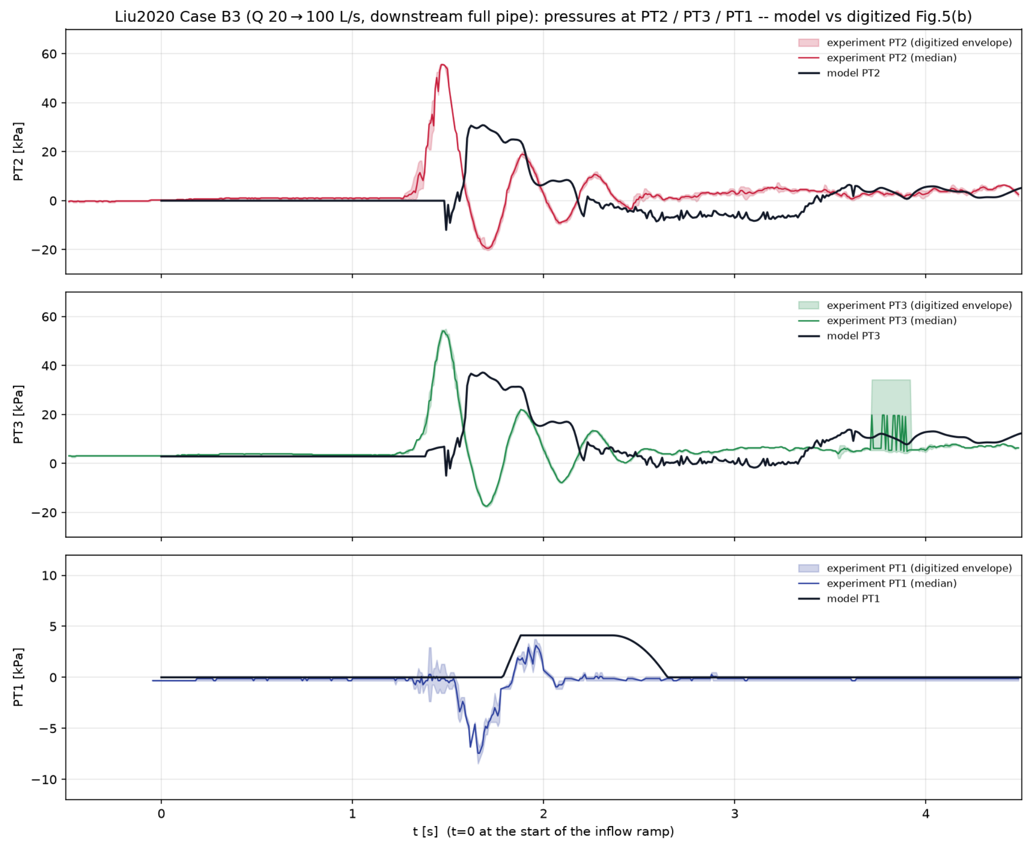

Figure 10: Case B3 (Q: 20→100 L/s, downstream surcharged): PT1–PT3 against time. Blue: digitized Fig. 5(b) of [5]; red: present model. The model selects the geysering branch; compressive peaks are acoustically under-resolved at the frozen $a_{wh} = 40$ m/s.

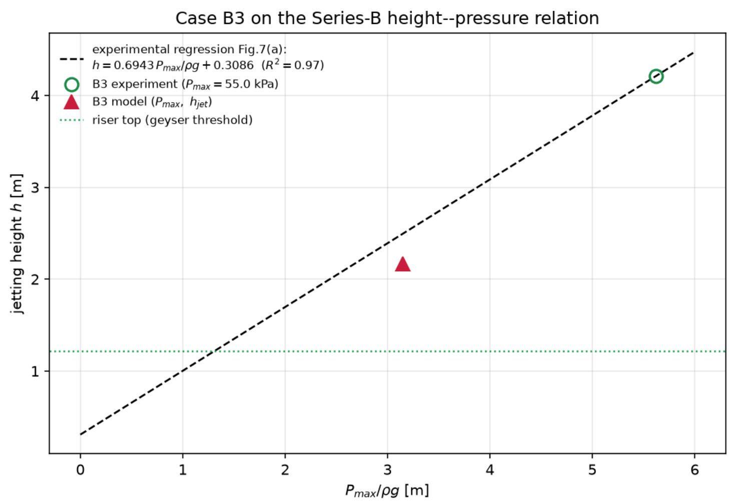

Figure 11: Case B3 on the eruption-height–peak-pressure diagram of Fig. 7(a) of [5]. Open symbols: experimental family; filled red circle: present model ($h = 2.17$ m, $P_{\max} = 31$ kPa at $a_{wh} = 40$ m/s).

For the phase-one comparison the pocket is omitted in the computation (the all-water variant assumed in Eq. (7) of [5]), while the sealed-pocket run is retained as a qualitative cushioning check. The analytic head at the chamber lid is

$$H_s(t) = h_{r0} + \frac{L_d}{gA_d}\frac{\mathrm{d}Q_u}{\mathrm{d}t}\left(1 - \cos\frac{2\pi t}{T}\right), \qquad T = 2\pi\sqrt{\frac{L_d A_r}{gA_d} + \frac{h_{r0}}{g}}, \tag{12}$$

with $T = 1.52$ s for C9.

Figure 12 compares the digitized PT2 record of Fig. 9 (converted to head at the chamber-top datum) against Eq. (12) and the all-water model over the first 5 s. The first peak is reproduced in timing and magnitude: measured $P_{1m} = 10.69$ kPa at $t = 0.50$ s, model 11.5 kPa at 0.54 s; the free surface reaches the shaft top at 0.73 s in the experiment and 0.93 s in the model. The oscillation period is $T = 1.45$ s in the paper (Eq. (8)) against 1.13 s in the model and 1.52 s in Eq. (12); the model is closer to the analytic value than to the measured period but still fast, consistent with the stiff one-dimensional tail-gate coupling. Steady heads after the first transient agree within ≈ 20% (PT2 final 8.79 vs 10.6 kPa). The sealed-pocket variant delays and cushions the first peak as in the experiment qualitatively, then drifts after ≈ 4 s when the wedge compresses through the reduced gas model—declared out of scope for phase two.

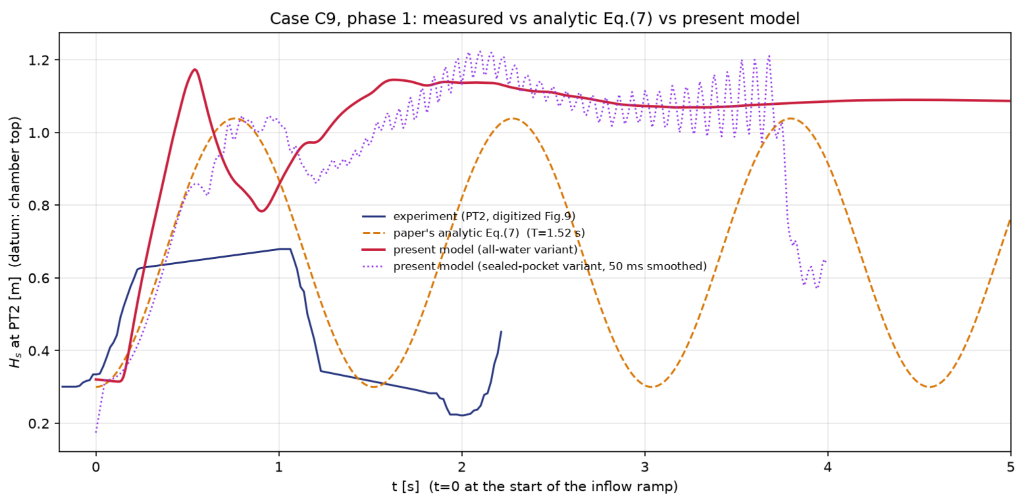

Figure 12: Case C9, phase 1 ($t < 5$ s): head H_s at PT2 (chamber-top datum). Blue: digitized Fig. 9 of [5]; orange dashed: Eq. (12); red: present model (all-water variant); purple dotted: sealed-pocket variant (50 ms smoothed). Pocket-driven geysers after $t \approx$ 6.5 s are not computed.

Taken together, the three cases show that the composite-domain formulation extends the frozen Campaign 1–2 solver to a prototype-motivated junction geometry: a downstream-boundary branch pair (A2/B3) and a transient mass-oscillation geyser (C9 phase 1) are reproduced at the level of regime selection, timing, and—where the physics is hydraulic rather than

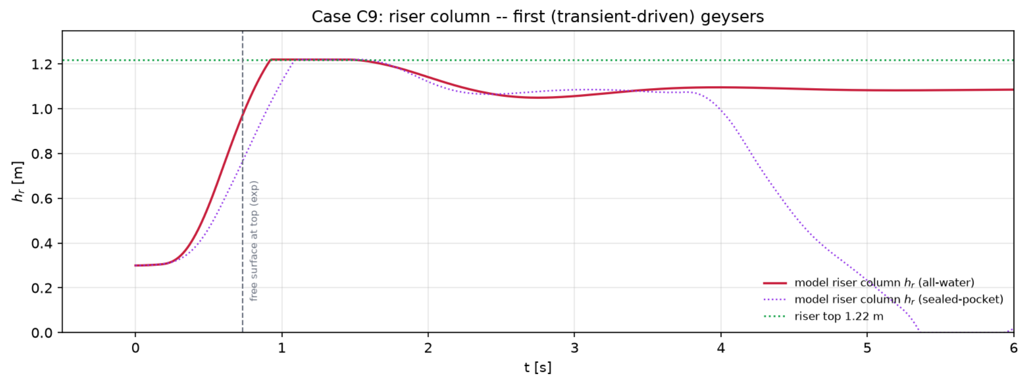

Figure 13: Case C9: shaft water-column height h_r for the all-water (solid) and sealed-pocket (dotted) variants. The dashed line marks the experimental time at which the free surface first reaches the shaft top in phase 1.

Table 3: Campaign 3 summary metrics. Acoustic peaks use the frozen $a_{wh} = 40$ m/s; no coefficient is retuned between cases.

Quantity	A2 (no geyser)	B3 (geyser)	C9 phase 1
Geyser?	no / no	yes / yes	yes / yes
PT2 steady or peak [kPa]	1.80 / 2.15	31 / 55	11.5 / 10.7
Peak or top-arrival time [s]	—	1.68 / 1.47	0.54 / 0.50
Shaft top reached [s]	—	≈ 1.65	0.93 / 0.73
Oscillation period [s]	0.30 (PT2–PT3)	—	1.13 / 1.45

Each cell: model / experiment.

acoustic—amplitude. The principal residual in this campaign is compressive under-resolution of short pressure spikes when the frozen elastic celerity is used, and the deliberate omission of pocket-transport physics after the pocket arrives at the chamber.

4. Discussion

4.1. Evidence across the three campaigns

The three campaigns test the model at different levels. Campaign 1 asks whether the computed gas and liquid fields reproduce the sequence within an individual event. Campaign 2 asks whether those fields select the correct branch across changes in shaft diameter and pocket volume. Campaign 3 asks whether the same reduced description remains informative when a junction chamber and downstream boundary participate in the transient. Considered together, the results show that conserving and transporting the gas phase is enough to recover the principal mechanism across all three apparatuses. They also show that correct branch selection does not guarantee accurate event timing, interface speed, or pressure peaks.

In Campaign 1, the wide and narrow towers occupy different experimental branches, and the model separates them correctly. For the wide tower, the computed maximum free-surface level is $0.66L$, compared with $0.63L$ in the experiment. The interface velocity is $0.41\sqrt{gD}$, compared with $0.39\sqrt{gD}$. The computed pressure plateau is also close to the measured level. These agreements indicate that the resolved pocket volume and hydrostatic resistance set the bulk response correctly. The remaining error is topological. The experiment contains a coherent gas pocket with a draining wall film, whereas the present grid-scale closure transports the shaft gas as a dispersed phase. It therefore misses the gradual free-surface creep and delays the pressure decay until breakthrough. After breakthrough, the local transducer state is repeatedly re-pressurized by the elastic horizontal water-column–pocket mode. The moving mean converts this oscillatory release into a gradual tail, but it does not cause the late onset.

The narrow tower exposes a different error. The model reproduces the pre-eruption stand ($0.84L$ versus $0.82L$), the pressure plateau, the rise to the rim, and the free-surface speed ($0.39\sqrt{gD}$ versus $0.44\sqrt{gD}$). However, the interface speed is only $0.71\sqrt{gD}$, compared with $1.43\sqrt{gD}$. The free surface reaches the rim early ($t^* = 3.70$ versus 3.90–3.98), whereas the interface reaches $0.85L$ late ($t^* = 4.44$ versus the credible high-interface cluster at 4.09–4.17). The error is therefore a stretched gas-release sequence, not a uniform shift of all events. This distinction is important because it localizes the deficiency to the transfer of gas through the shaft mouth and the pocket-fed core rather than to the initial pocket arrival alone.

Campaign 2 shows how this timing and exchange error affects regime classification. The model captures 5 of the 7 filmed Series-B cases and 39 of the 63 configurations in the full parameter sweep. It reproduces the geysering end member, all wide-riser and small-pocket non-geysering end members, and the measured branch differentiation of pipe-end pressure. The 24 sweep disagreements are all experimental geysers that remain below the rim in the computation. No computed case produces a false geyser. This one-sided pattern is mechanistically consistent with insufficient net water- column lift near the transition band. It should not be described as "conservative" for hazard screening, because missed eruptions are non-conservative from a safety perspective.

Campaign 3 provides a complementary boundary-condition test. Changing only the downstream condition reverses the outcome between A2 and B3, and the model reproduces that reversal. It also places the B3 jet height on the experimental pressure–height trend. The frozen wave speed under-resolves the short pressure peak, yielding 30.84 kPa rather than 55.03 kPa. For C9, the computed first pressure peak is 11.49 kPa at 0.54 s, compared with 10.69 kPa at 0.50 s. The predicted oscillation period is shorter than measured, and later pocket-driven geysers lie outside the composite one-dimensional representation. Thus, the chamber reduction retains the first transient response but not the full later-time gas-pocket dynamics.

4.2. Where the reduced model succeeds

The strongest result is the recovery of branch physics without prescribing a gas-release history. In each campaign, the pocket pressure evolves from the conserved gas mass and computed volume. Gas transport into the shaft and water displacement then follow from the coupled solution. This structure allows the model to distinguish benign venting from eruption and to respond to changes in riser diameter, pocket volume, and downstream boundary. A liquid-only model with a prescribed pressure or discharge could reproduce a selected trace, but it would not provide the same independent regime test.

The results also identify which observables are robust at one-dimensional resolution. Slowly varying pressure plateaus, quasi-static water-column stands, maximum free-surface levels, and the first mass-oscillation peak are captured more reliably than interface breakthrough, annular-film motion, or short acoustic spikes. These robust quantities are governed mainly by gas inventory, liquid-column inertia, and hydrostatic balance. The less robust quantities depend on local three-dimensional exchange or unresolved time scales at the junction and shaft wall.

4.3. Closure limitations revealed by the experiments

The dominant limitation is the shaft-mouth closure. In the narrow-tower configuration, a crown-referenced pressure datum, a pocket-fed gas core, and a Wallis-type flooding limit are activated by the resolved driven regime. These elements use geometry and literature-scale constants rather than case-specific calibration. Nevertheless, they remain a regime-conditioned closure rather than a unified cross-sectional junction solution. The wide tower uses an invert-referenced datum, while the narrow tower requires the physical crown connection. A future junction model should transmit the vertical pressure distribution across the mouth instead of reducing it to a single scalar datum.

The second limitation is the coupling between cavity propagation and vent feed. The frozen dissipative cavity speed places the initial arrivals within the observed scatter. The same reduced feed then slows the pocket-fed core and delays breakthrough. Separating the front speed from the pressure-driven gas flux through the mouth would allow the two constraints to be satisfied independently.

The third limitation is the volume-neutral counter-current exchange used at the tee. This exchange preserves pocket pressure during balanced venting and is needed to produce the B-H1 geyser. In reservoir-fed, wider-riser cases, however, it cancels part of the water-column displacement produced by entering gas. Removing the exchange restores lift in those cases but also removes the B-H1 pressure rebuild. The two branches therefore require a flux-limited exchange law based on local counter-current capacity, not a fixed equality of gas and liquid volumes.

Two additional limits follow from dimensional reduction. A dispersed two- fluid state does not preserve the coherent Taylor-pocket topology and its draining film in the wide tower. A reduced chamber and softened acoustic celerity do not resolve the shortest B3 pressure spike or the later arrival and venting of pockets in C9. These discrepancies should be addressed by targeted closure development or local multidimensional coupling. They should not be hidden by adjusting a global parameter to individual tests.

4.4. Implications and scope of use

The present evidence supports using the framework to study mechanisms and to screen system-scale scenarios away from the regime boundary. It can track gas inventory, pocket compression, branch selection, and the bulk pressure and water-column response within one computational description. The evidence does not support using the current formulation as a stand-alone safety classifier near the boundary. Its Campaign 2 errors are missed geysers, and the local pressure peak in B3 is sensitive to acoustic representation.

The reported accuracy applies to the frozen numerical configuration and the documented, regime-conditioned junction closures. It does not establish universality across shaft shapes, manhole covers, network interactions, or prototype Reynolds and Weber numbers. The next validation step should therefore target an independent apparatus without modifying the closure parameters. Priority measurements are simultaneous gas flux, liquid return flow, pocket pressure, and interface topology at the shaft mouth. Such data would distinguish an exchange-law error from a shaft-entrainment error and would make the next closure revision falsifiable.

5. Conclusions

This study assessed a gas-conserving one-dimensional mixed-flow framework against three published geysering campaigns. The model represented the shaft and horizontal conduit with coupled two-fluid states and computed the gas release rather than prescribing it. Across the ventilation-shaft tests, it selected both the non-geysering and geysering branches and reproduced the principal pressure and water-column scales. In the wide tower, the pressure plateau was reproduced, but blowdown was delayed until breakthrough and relaxed too gradually. The narrow-tower comparison revealed a stretched release sequence: shaft arrival was early, whereas interface breakthrough was late and the computed interface velocity was approximately one half of the measured value.

Across the horizontal-pipe/vertical-riser campaign, the model classified 5 of 7 filmed cases and 39 of 63 parameter-sweep configurations. All 24 sweep errors were missed experimental geysers near the transition band, with no false geysers. In the junction-chamber campaign, it reproduced the A2/B3 branch reversal and the first C9 pressure transient, but under-resolved the short B3 pressure peak and did not represent later pocket-driven events.

The combined evidence separates a useful system-scale capability from its current limits. Resolved gas conservation and transport recover the dominant regime mechanism, bulk pressure levels, and water-column response.

Remaining errors concentrate in shaft-mouth exchange, coherent-film topology, and acoustic/chamber reduction. The framework is therefore suitable for mechanistic assessment and qualified screening away from the boundary. A unified, flux-limited junction and entrainment closure is required before the model can be relied upon for safety-critical classification of marginal geysering cases.

Data Availability Statement

The data and materials supporting the findings of this study are available from the corresponding author upon reasonable request. A persistent archive identifier will be added if the project repository is deposited before submission.

Declaration of Generative AI Use

Generative AI tools were used to assist language revision and manuscript organization. The authors critically reviewed and edited all generated material and take full responsibility for the content of the manuscript.

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